pfi

Python for Fixed Income

COURSE 7 · NLP, RAG, AND LLMS

Zero-Shot Classification against TF-IDF

WEEK 2 · DAY 2

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This course is educational content created in a personal capacity. Nothing here is investment advice or a recommendation to buy or sell any security. All portfolio data is synthetic.

Three models, one set of rows

TF-IDF (day 3)

Fitted on 3,650 labelled sections.

Zero-shot

No labelled section: eight descriptions the course wrote.

Embeddings

MiniLM's 384 numbers, then a logistic regression.

Fitted on 3,650 sections from 2019 to 2023; scored on 741 from 2024; 735 from 2025 closed.

Real documents, described

An 8-K section's label is the filer's item; a model's output is the item it assigns

Allowed names	Never called
the item the model assigns	a default signal
the model's probability or score in this file	a distress flag or a warning
the filer's item	a credit event or credit view

Item 5.02 sections name people: counted by item only, never printed.

Natural language inference, used as a classifier

Premise

The 8-K section, its first 120 words.

Hypothesis, one per label

This section of a company's current report is about an event that accelerates a debt obligation.

12,750,594

parameters

8

passes per section

6

files, 26,457,473 bytes

The pipeline returns the labels sorted by score, largest first.

Where the transformer comes from

The paper	"Attention Is All You Need": arXiv 1706.03762, submitted 2017-06-12
Authors	Vaswani, Shazeer, Parmar, Uszkoreit, Jones, Gomez, Kaiser, Polosukhin
Its abstract	"the Transformer, based solely on attention mechanisms"

The course uses the model as published. The paper itself was not read.

Source, read 8 October 2026: arxiv.org/abs/1706.03762 (its arXiv API record)

The label wording matters

5 of 8

the course's wording

4 of 8

the SEC's item titles

128

passes of the model

One training section of each item, drawn with the seed. Eight is too few to choose on; it shows the words move the result.

Label wording is fixed on training rows, never on the rows that score the model.

A cached output, checked live

3,675

seconds: the tool on one thread

25 of 25

the same item live and from the file

True

live scores equal the file's

```
tolerance = 1e-4 if torch.get_num_threads() == 1 else 1e-3
# one thread: within 1e-6 of the file; default threads moved scores by up to about 3e-4
```

A live zero-shot score is never printed: scores print from the committed file.

Three models on the validation rows

	TF-IDF	Zero-shot	Embedding
recall 1.01	0.9100	0.9400	0.8700
recall 1.03	0.8800	0.6400	0.8700
recall 2.03	0.8300	0.0500	0.7300
recall 2.04	0.9200	0.3300	0.8500
macro F1	0.9271	0.5951	0.8882

Zero-shot assigns 1.01 to 0.3698 of the sections. 50 live embeddings equal the committed rows within 1e-6: True.

The logistic regression is fitted on float64 copies: on float32 the solver stopped short.

Where the three models agree

0.6302

TF-IDF and zero-shot

0.9312

TF-IDF and embedding

0.6289

zero-shot and embedding

All three assign the same item to 452 of 741 sections; on those it is the filer's item 0.9823 of the time.

Agreement is not accuracy: two models can agree and both be wrong.

A sentiment pipeline, named and not run

```
SUPPORTED_TASKS["text-classification"]["default"]   # read, nothing downloaded
# distilbert/distilbert-base-uncased-finetuned-sst-2-english, revision 714eb0f
```

Its model card: "fine-tuned on SST-2". The SST-2 card: "11,855 single sentences extracted from movie reviews".

A label learned from film reviews means nothing for an 8-K section.

Sources, read 8 October 2026: the transformers 5.19.0 source; huggingface.co model and dataset cards

Excel side by side

Excel side by side (tested in Excel)

`=INDEX($B$1:$I$1,MATCH(MAX(B2:I2),B2:I2,0))`	the zero-shot item on 741 of 741 rows
`=SUMPRODUCT(--(K2:K742=J2:J742))/ROWS(K2:K742)`	0.6302, within 1e-12
`=COUNTIFS(K2:K742,"2.04",J2:J742,"2.04")`	33

No zero-shot or sentence model in Excel: the scores come from PyTorch.

Save the day: git diff results.csv

```
git diff results.csv
+zero_shot,validation,0.5951,0.6262,0.6400,0.3300,741
+embedding_logistic,validation,0.8882,0.8934,0.8700,0.8500,741
```

pytest: 40 passed. A log with 4 decimals and LF endings diffs cleanly.

The AI check

The docstring is the prompt. The code that comes back runs, and its number looks fine.

1

Docstring

Units, conventions, and what the function must never do: what you asked for.

2

Code back

Written for the course in the style of an AI answer. Read it.

3

Test first

Assert what the docstring promises. Run it: it fails.

4

Fix

Find the rule it broke, fix it, run the test again.

Today: sorted scores under fixed labels: all 741 sections assigned 1.01, macro F1 0.0297.

Code that runs is not the same as code that is right.

What you can do now

- One inference pass per label
- Read every score by its label's name
- Fix label wording on training rows
- Check a cached output with a live sample
- Agreement is not accuracy

Week 2, Day 3 Clustering 10-K risk factors